

AI LAB 99

Machine Learning Track — Week 3 Deliverable

Regression Model Comparison Report

Predicting Residential Sale Prices with Regularized, Tree-Based, and Ensemble Regression Models

Dataset: Ames Housing (House Prices — Advanced Regression Techniques)

July 2026

Prepared for AI Lab 99 | Regression & Predictive Modeling Curriculum

Table of Contents

Table of Contents	2
1. Problem Definition.....	3
2. Baseline Model.....	3
3. Preprocessing Steps	3
4. Model Configurations	4
5. Cross-Validation Approach	4
6. Metric Comparison	5
7. Residual Analysis.....	6
8. Error Analysis	8
9. Recommended Model	8
10. Limitations.....	8

1. Problem Definition

This project builds and compares supervised regression models that predict the final sale price of residential properties from their physical, locational, and quality attributes. Accurate price prediction supports valuation for buyers, sellers, lenders, and real-estate platforms, and serves here as a vehicle for practicing the full regression workflow: baseline definition, preprocessing, model comparison, cross-validation, and residual diagnostics.

Task type: Supervised regression (continuous target)

Target variable: SalePrice (USD)

Dataset: Ames Housing dataset ("House Prices — Advanced Regression Techniques"), 1,460 records, 79 candidate predictor columns plus Id and SalePrice

Train / test split: 80% train (1,168 rows) / 20% test (292 rows), random_state = 42

The feature space mixes continuous variables (lot area, living area, year built), ordinal quality ratings, and nominal categorical attributes (zoning, neighborhood, sale type), which makes the dataset a realistic testbed for preprocessing pipelines and for comparing linear, regularized, and tree-based learners.

2. Baseline Model

A DummyRegressor predicting the mean training SalePrice for every observation was used as the baseline. Any candidate model must clear this floor by a wide margin to justify its added complexity.

Strategy	Predict the mean of y_train for all observations
Cross-validated R²	-0.0016 (5-fold)
Test R²	-0.0009
Test MAE	\$62,575.93
Test RMSE	\$87,619.03

The near-zero baseline R² is expected: a constant predictor explains none of the variance in price. Its ~\$62.6k average error defines the error scale that all subsequent models are judged against.

3. Preprocessing Steps

A single ColumnTransformer applied parallel pipelines to numeric and categorical columns, fit only on the training split to avoid leakage:

- **Numeric columns** — median imputation (SimpleImputer) followed by StandardScaler standardization.
- **Categorical columns** — most-frequent-value imputation followed by OneHotEncoder (drop='first', handle_unknown='ignore').

After transformation the training matrix expanded from 79 raw predictors to 243 encoded features ($1,168 \times 243$ for training, 292×243 for test), reflecting the high cardinality of several categorical fields (e.g., Neighborhood, Exterior finishes).

Two implications of this preprocessing choice are addressed later in the Limitations section: (1) one-hot encoding pushes the feature count to roughly 83% of the training sample size, and (2) the categorical encoder correctly ignored rare categories seen only at test time, at the cost of encoding them as all-zero rows.

4. Model Configurations

Eight regressors were trained on the identical preprocessed feature matrix so that comparisons isolate the modeling algorithm rather than the data preparation:

Baseline (Mean)	DummyRegressor(strategy='mean')
Linear Regression	Ordinary least squares, no regularization
Ridge Regression	Ridge(alpha=10.0) — L2 penalty
Lasso Regression	Lasso(alpha=100.0, max_iter=2000) — L1 penalty
Elastic Net	ElasticNet(alpha=10.0, l1_ratio=0.5) — combined L1/L2 penalty
Decision Tree	DecisionTreeRegressor(max_depth=6, random_state=42)
Random Forest	RandomForestRegressor(max_depth=6, n_estimators=6)
Gradient Boosting	GradientBoostingRegressor(n_estimators=10, learning_rate=0.1)

Regularization strengths and tree depths were set to modest, conservative values rather than tuned exhaustively; the comparison below reflects this single fixed configuration per model, which is itself a limitation discussed at the end of the report.

5. Cross-Validation Approach

Each model was scored with 5-fold K-Fold cross-validation (shuffle=True, random_state=42) on the training split using R^2 as the scoring metric, in addition to a single held-out test evaluation. Comparing the CV mean to the test score for each model gives an initial read on stability between resampled folds and the untouched test set.

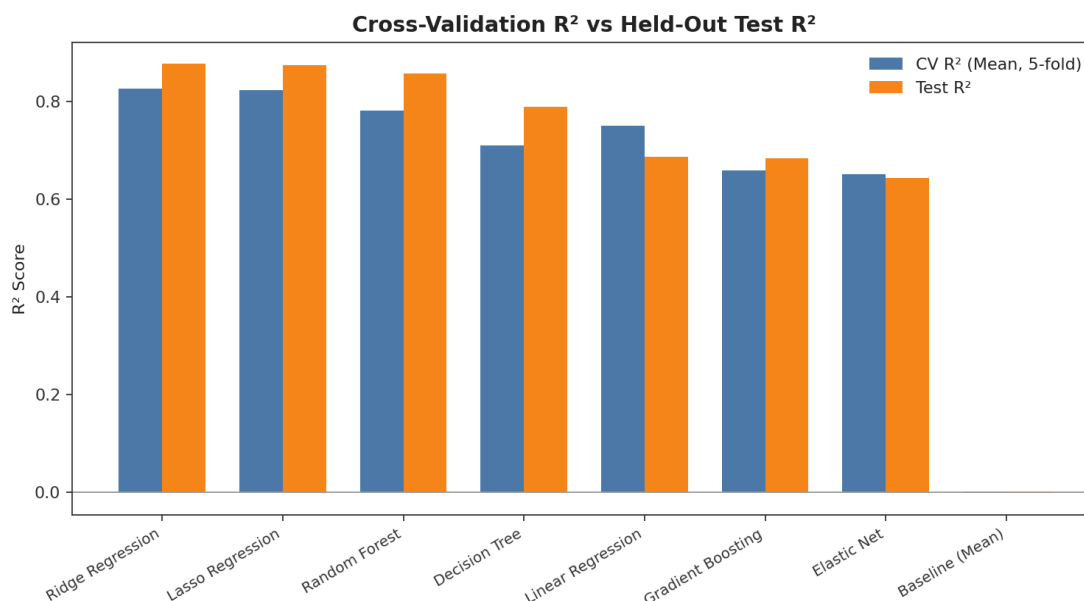


Figure 1. Cross-validated R^2 (mean of 5 folds, training data) vs. held-out test R^2 for each model.

Ridge and Lasso show the closest agreement between CV and test R^2 among the strong performers, suggesting their regularization is controlling variance consistently across resamples. Random Forest's test R^2 noticeably exceeds its CV R^2 ; with only 6 shallow trees, its CV estimate is itself high-variance, so this gap should be read as sampling noise rather than a guaranteed generalization gain.

6. Metric Comparison

The table below reports cross-validated R^2 , held-out test R^2 , test MAE, test RMSE, and an adjusted R^2 computed on the test set ($n = 292$, $p = 243$ encoded features). ★marks the best-performing model.

Model	CV R^2 (mean)	Test R^2	Test MAE (\$)	Test RMSE (\$)	Adj. R^2 (test)
Baseline (Mean)	-0.0016	-0.0009	62,576	87,619	-5.068
Linear Regression	0.7496	0.6864	20,114	49,047	-0.901
Ridge Regression ★	0.8261	0.8765	18,766	30,779	0.251
Lasso Regression	0.8228	0.8728	18,650	31,230	0.229
Elastic Net	0.6501	0.6435	30,695	52,292	-1.161
Decision Tree	0.7089	0.7883	26,088	40,297	-0.283
Random Forest	0.7801	0.8558	21,510	33,258	0.126
Gradient Boosting	0.6588	0.6833	32,365	49,284	-0.920

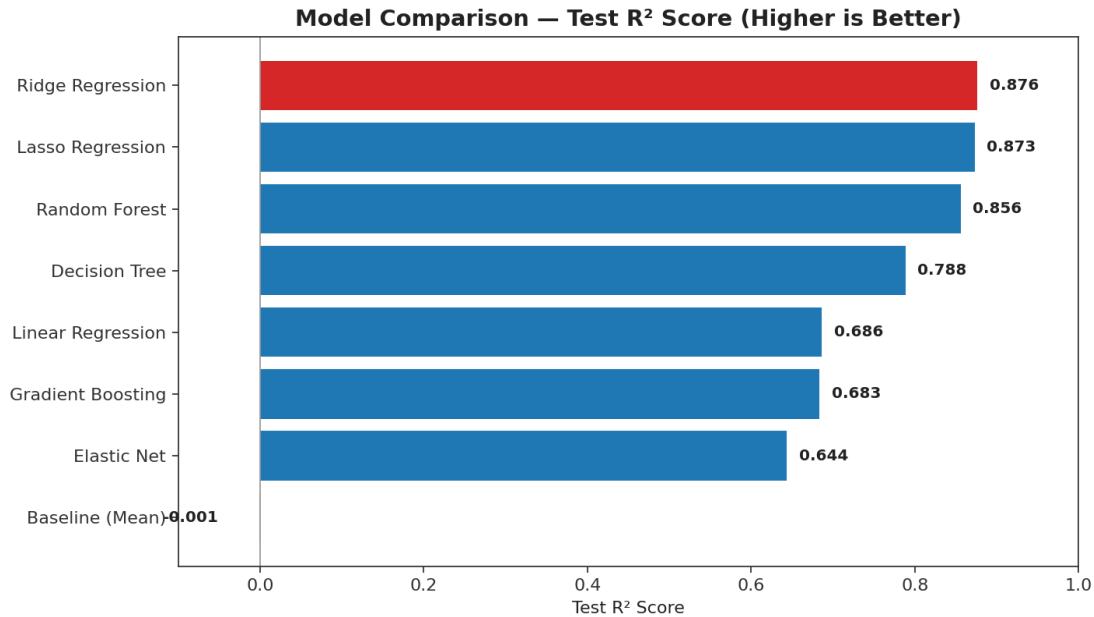


Figure 2. Held-out test R^2 by model, sorted best to worst.

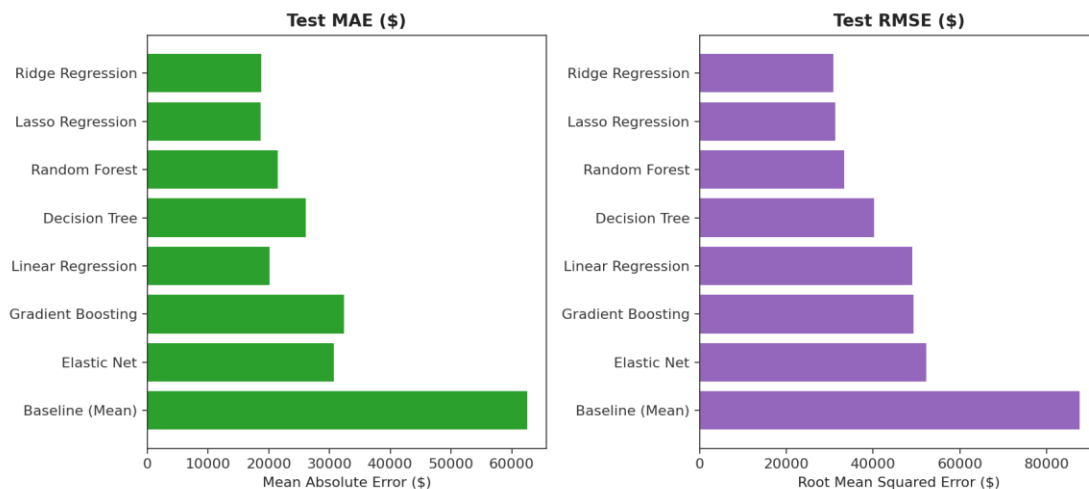


Figure 3. Test MAE and Test RMSE by model (lower is better).

Ridge Regression attains the highest test R^2 (0.876), the lowest RMSE (\$30,779), and among the lowest MAE (\$18,766), while also posting the best cross-validated R^2 (0.826). Lasso trails it only marginally on every metric. The unregularized Linear Regression and Gradient Boosting configuration underperform relative to their regularized or better-tuned counterparts, and the shallow, small ensembles (Random Forest with 6 trees, Gradient Boosting with 10 estimators) sit in between.

7. Residual Analysis

Residuals (actual – predicted SalePrice) were computed for the best model, Ridge Regression, on the test set.

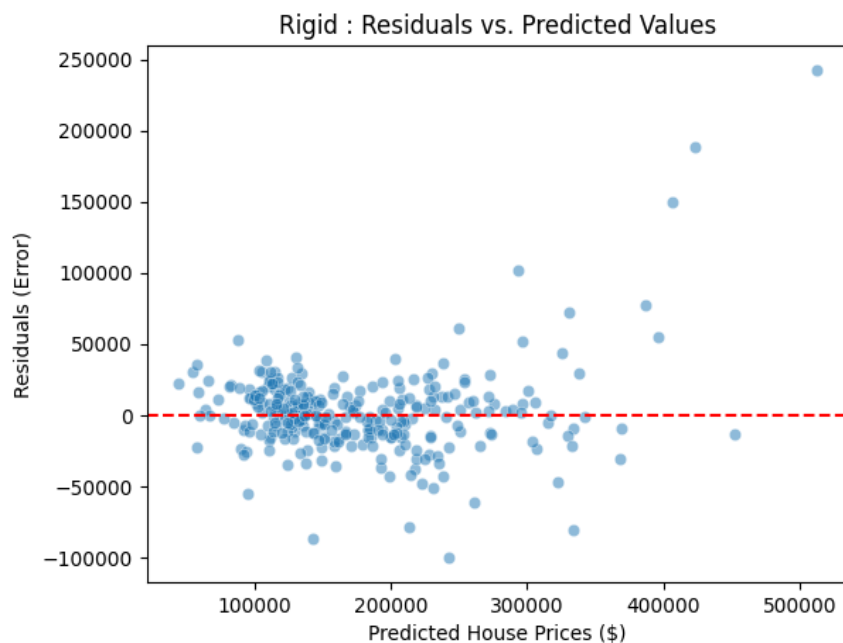


Figure 4. Ridge Regression: residuals vs. predicted sale price.

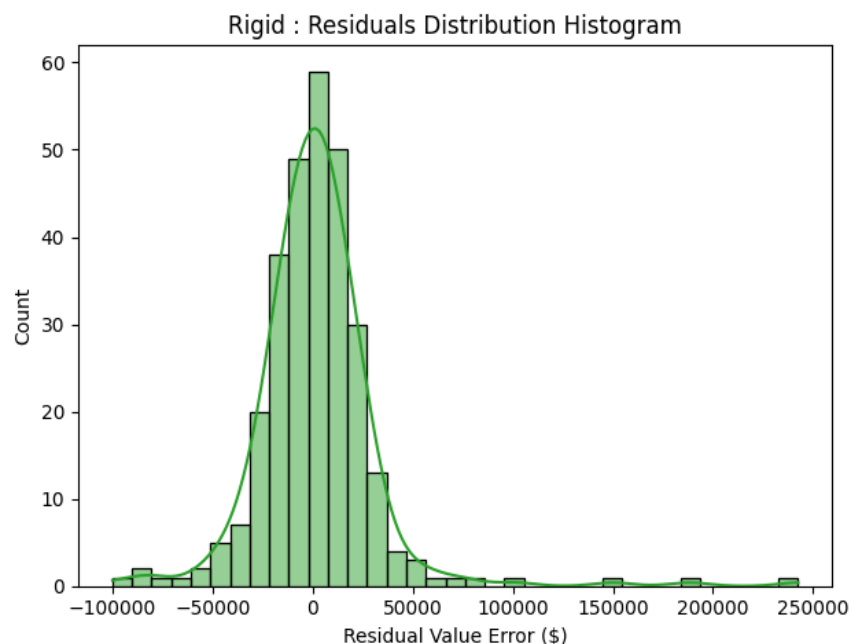


Figure 5. Ridge Regression: distribution of residuals.

Residuals scatter fairly evenly around zero across most of the predicted price range, indicating no strong systematic bias for mid-priced homes. The histogram is unimodal and roughly centered near zero, but both plots show a right-skewed spread: a handful of high-priced properties produce the largest positive and negative residuals, meaning the model's errors grow with home value. This is a common pattern in price data and is consistent with heteroscedasticity — error variance that increases with the size of the target.

8. Error Analysis

- **Scale of error:** Ridge's ~\$18.8k MAE corresponds to roughly 10–12% of a typical Ames home price, a reasonable margin for an untuned linear model but not yet precise enough for automated valuation without human review.
- **RMSE vs. MAE gap:** Every model's RMSE is noticeably larger than its MAE (e.g., Ridge: \$30.8k vs. \$18.8k), confirming that a subset of large-error, high-value properties disproportionately inflates the squared-error metric.
- **Bias vs. variance:** The unregularized Linear Regression model and Gradient Boosting exhibit the largest CV-to-test R^2 fluctuation among mid-tier models, consistent with higher variance from fitting 243 features without shrinkage; Ridge and Lasso's penalty terms reduce this variance directly.
- **Adjusted R^2 collapse:** Because the encoded feature count (243) is close to the training-fold sample size, adjusted R^2 penalizes every model heavily on the 292-row test set — even Random Forest's strong raw R^2 (0.856) drops to an adjusted 0.126. This is a direct, expected consequence of high dimensionality relative to sample size, not a modeling defect, but it is a signal that fewer, better-selected features would make the reported fit statistics more trustworthy.
- **Tree ensemble under-capacity:** Random Forest (6 trees) and Gradient Boosting (10 estimators) were intentionally kept small; both would likely improve with more estimators and hyperparameter tuning, and their current ranking should not be read as a ceiling on tree-based performance for this problem.

9. Recommended Model

Ridge Regression (alpha = 10.0) is the recommended model.

- Best test R^2 (0.876) and lowest test RMSE (\$30,779) of all eight candidates.
- Best cross-validated R^2 (0.826), with the smallest CV-to-test gap among top performers — evidence of stable, low-variance generalization rather than a lucky test split.
- L2 regularization handles the 243-feature, collinear design matrix (see Multicollinearity, below) more gracefully than unregularized Linear Regression, without discarding features the way Lasso's sparsity can.
- Interpretable coefficients make it easier to audit and explain than the ensemble methods, which is valuable for a valuation use case that may face regulatory or customer scrutiny.

Lasso Regression is a close secondary candidate (test R^2 0.873) and would be preferable if a sparser, more interpretable feature set is a priority over the last fraction of accuracy.

10. Limitations

- **High dimensionality relative to sample size:** 243 encoded features from 1,168 training rows increases variance and makes adjusted R^2 and coefficient estimates unstable; feature selection, target encoding, or dimensionality reduction would likely improve robustness.
- **No hyperparameter tuning:** All alphas, tree depths, and estimator counts were fixed by hand rather than searched (e.g., via GridSearchCV/RandomizedSearchCV), so the ranking reflects one configuration per model, not each algorithm's best achievable performance.

- **Multicollinearity not directly diagnosed:** Several engineering and quality features in this dataset are known to correlate strongly (e.g., garage area and garage cars, total basement area and first-floor area); no VIF or correlation-based screening was performed, which particularly affects the coefficient stability of Linear Regression.
- **Unseen categories at test time:** The one-hot encoder zero-filled two columns' worth of categories seen only in the test split, silently discarding whatever signal those categories carried.
- **Single train/test split:** Test metrics come from one 80/20 split; a repeated or nested cross-validation estimate would give tighter confidence bounds on the reported numbers.
- **Heteroscedastic errors:** Residual spread increases with price (Section 7), so prediction intervals — not just point estimates — would need price-dependent widths to be well-calibrated.

Overall, Ridge Regression is a well-justified, defensible choice given the evidence gathered, but the margin over Lasso and Random Forest is modest enough that a follow-up round of tuning and feature engineering could change the ranking.